

Weighted collision bounds for triple minima and six-window swaps

Research proof candidate; independent review pending

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Let $\rho_1, \dots, \rho_M$ be permutations of $[N]$, with arbitrary nonnegative weights μ_b summing to one. Repetitions are allowed. Put $\kappa = \sum_b \mu_b^2$ and $m = N - 1$. The number $1/\kappa$ is the effective support size used below; the actual support may be much larger.

Lemma 1 (Averaging the centered Gram collision term). *For each pivot i , let $x_{b,i} \in \{0, 1\}^m$ have coordinates $\mathbf{1}\{\rho_b(i) < \rho_b(j)\}$ for $j \neq i$, and put*

$$P = I - m^{-1} \mathbf{1}\mathbf{1}^T, \quad R_i = \sum_b \mu_b (Px_{b,i})(Px_{b,i})^T.$$

If $N \geq 3$, some pivot i satisfies

$$\|R_i\|_F \geq \frac{m}{\sqrt{60}} \sqrt{\kappa}.$$

Proof. All the weights are nonnegative, so the Gram expansion gives

$$\|R_i\|_F^2 = \sum_{b,c} \mu_b \mu_c \langle Px_{b,i}, Px_{c,i} \rangle^2 \geq \sum_b \mu_b^2 \|Px_{b,i}\|^4.$$

For fixed b , as i ranges over the labels, the number k of ones in $x_{b,i}$ ranges over $0, 1, \dots, m$ exactly once. Its centered squared norm is $k(m - k)/m$. Therefore

$$\frac{1}{N} \sum_i \|Px_{b,i}\|^4 = \frac{1}{m+1} \sum_{k=0}^m \frac{k^2(m-k)^2}{m^2} = \frac{(m-1)(m^2+1)}{30m} \geq \frac{m^2}{60}.$$

Averaging the preceding Gram inequality over pivots proves the lemma. $\square$

Theorem 2. *Define*

$$\epsilon = \max_{i,j,k \text{ distinct}} \left| \Pr(\rho(i) < \rho(j), \rho(i) < \rho(k)) - \frac{1}{3} \right|.$$

Then

$$\epsilon \geq \frac{\sqrt{\kappa}}{\sqrt{60}} - \frac{2}{3\sqrt{N-1}}.$$

For $\delta(a, b, c) = \frac{1}{2} - \frac{3}{2} \Pr(\rho(b) < \rho(a), \rho(b) < \rho(c))$, let η be the maximum absolute change in the sum of four consecutive triple scores under a middle-pair swap in a six-label window. For $N \geq 6$ one has

$$\eta \geq \frac{3}{2\sqrt{60}}\sqrt{\kappa} - \frac{4}{\sqrt{N-1}} - \frac{128}{N}.$$

In particular, if $N \geq C/\kappa$ for a sufficiently large absolute constant C , then some such swap has magnitude at least $c\sqrt{\kappa}$.

Proof. The uncentered Gram matrix $G_i = \sum_b \mu_b x_{b,i} x_{b,i}^T$ has diagonal $p_{ij} = \Pr(\rho(i) < \rho(j))$ and off-diagonal triple-minimum probabilities. Thus

$$G_i = \frac{1}{3} \mathbf{1}\mathbf{1}^T + D_i + E_i, \quad |(D_i)_{jj}| \leq 2/3, \quad (E_i)_{jj} = 0, \quad |(E_i)_{jk}| \leq \epsilon.$$

Multiplication on both sides by P removes the constant matrix, giving $\|R_i\|_F \leq (2/3)\sqrt{m} + \epsilon m$. Combine with Lemma ?? to obtain the first assertion.

For the second assertion, apply the small-swap coboundary lemma from the companion note *Deterministic six-window discrepancy for short permutation families*. It supplies an antisymmetric array a_{ij} with $|a_{ij}| \leq 3/2$ such that

$$|\delta(j, i, k) - a_{ij} - a_{ik}| \leq \tau := \eta + 128/N.$$

Consequently, for every pivot i ,

$$G_i = A_i + D_i + E_i, \quad (A_i)_{jk} = \frac{1}{3} - \frac{2}{3}(a_{ij} + a_{ik}),$$

where the formula for A_i also specifies its diagonal,

$$(D_i)_{jj} = p_{ij} - \frac{1}{3} + \frac{4}{3}a_{ij}, \quad |(D_i)_{jj}| \leq \frac{8}{3},$$

and E_i has zero diagonal and off-diagonal entries at most $2\tau/3$ in absolute value. Every summand of the additive matrix A_i has a constant vector on one side, so $PA_iP = 0$. It follows that

$$\|R_i\|_F \leq \frac{8}{3}\sqrt{m} + \frac{2}{3}\tau m.$$

Use the pivot supplied by Lemma ?? and rearrange to obtain the second inequality. $\square$

Restriction and repeated blocks. The result applies after restricting all permutations to any common subset of labels, with exactly the same weights and the same κ . If two restrictions coincide, they may be left as repeated blocks; merging them only increases κ and strengthens the bound. This makes greedy packing of disjoint six-label windows possible whenever a fixed positive fraction of N remains and $N \geq C/\kappa$.